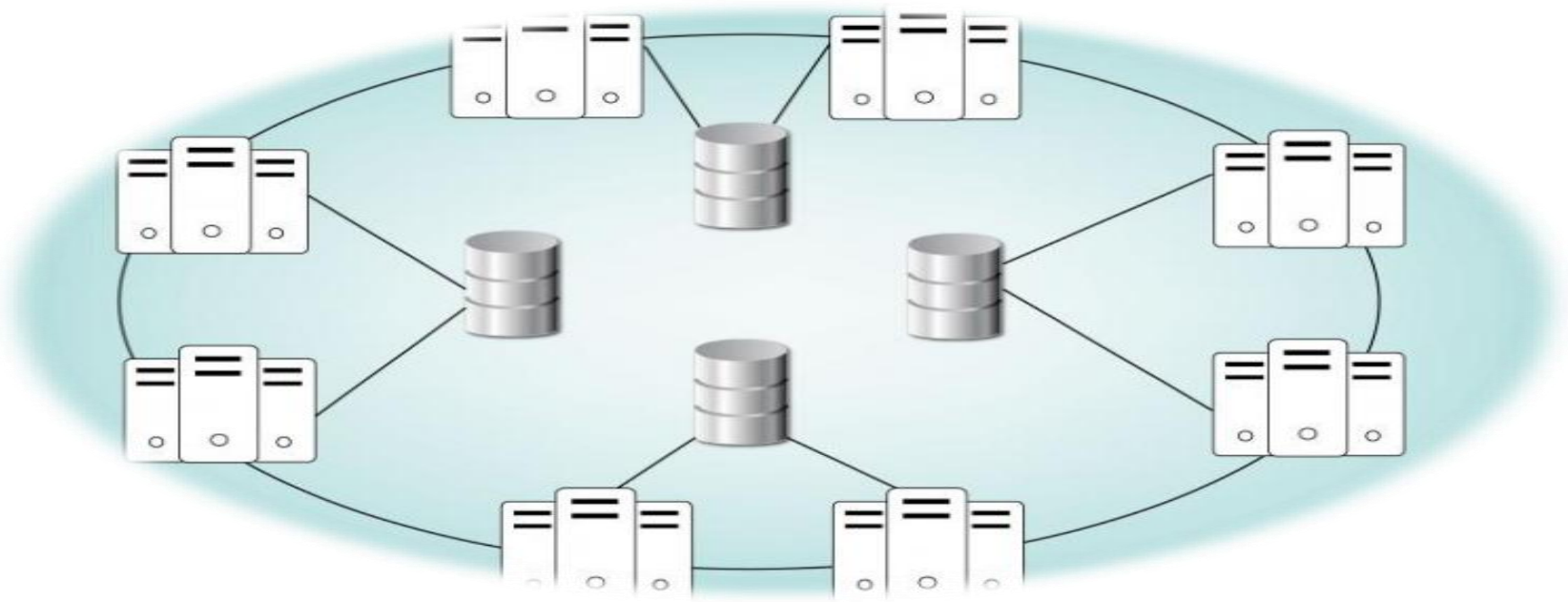
Information Technology Institute

ITI

Introduction to Computer Networks & Cyber Security

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Distributed Systems &Cloud Computing

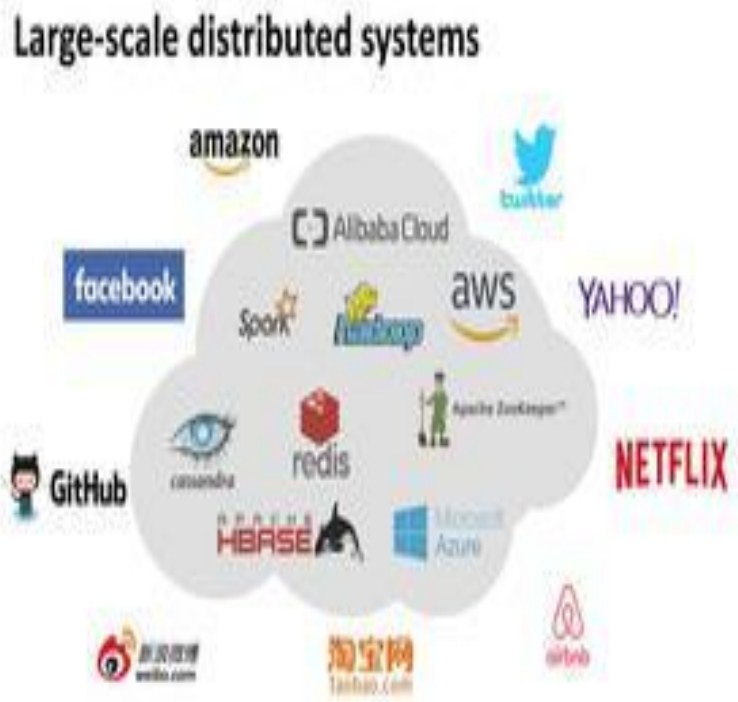


What is Distributed Systems ?

Distributed Systems

-A distributed system is a network of independent computers that work together to achieve a common goal, appearing to users as a single coherent system.

These systems can be located in the same physical location or spread across vast distances, enabling them to function as a unified entity despite their distributed nature



Centralized system VS Distributed system

- **Centralized system:** State stored on a **single computer**
  - Simpler
  - Easier to understand
  - Can be faster for a single user

- **Distributed system:** State divided over **multiple computers**
  - More robust (can tolerate failures)
  - More scalable (often supports many users)
  - More complex

Why build a distributed system?

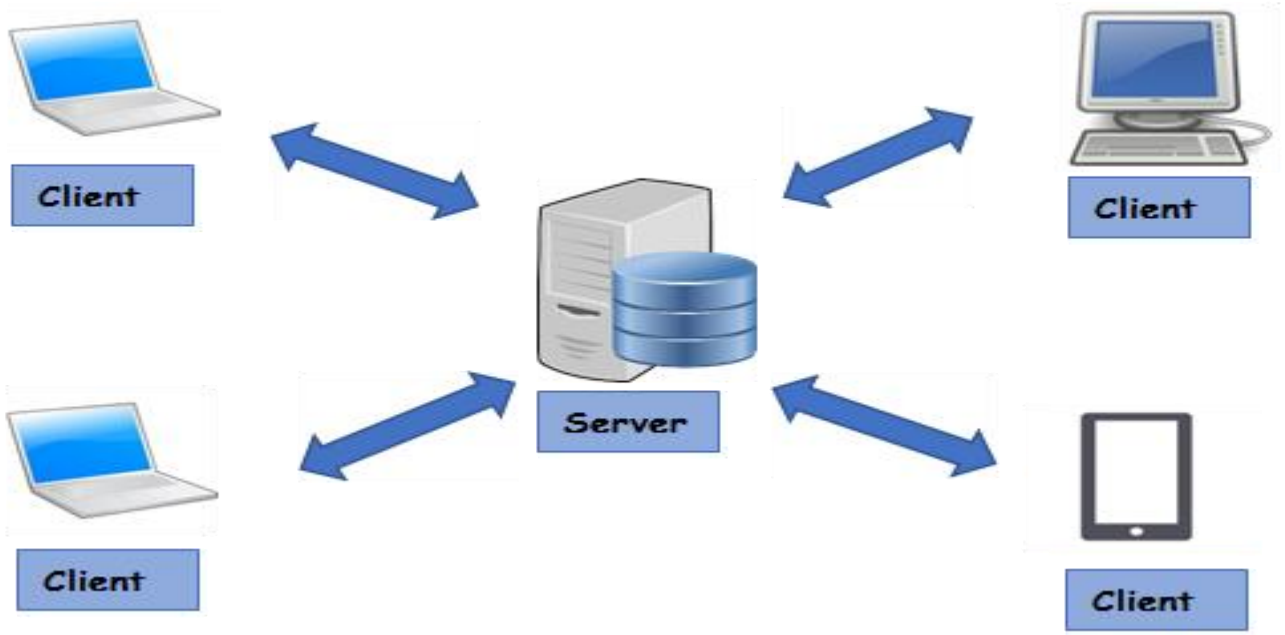
- **One interface** to the end-user.
- **Performance**
  - maximize resources and information while preventing failures
- **Reliability**
  - if one system fails, it won't affect the availability of the service
- **Dependency on cloud**
- **Scaling**

Distributed System Architectures:

Four architecture types :

1- Client-server:

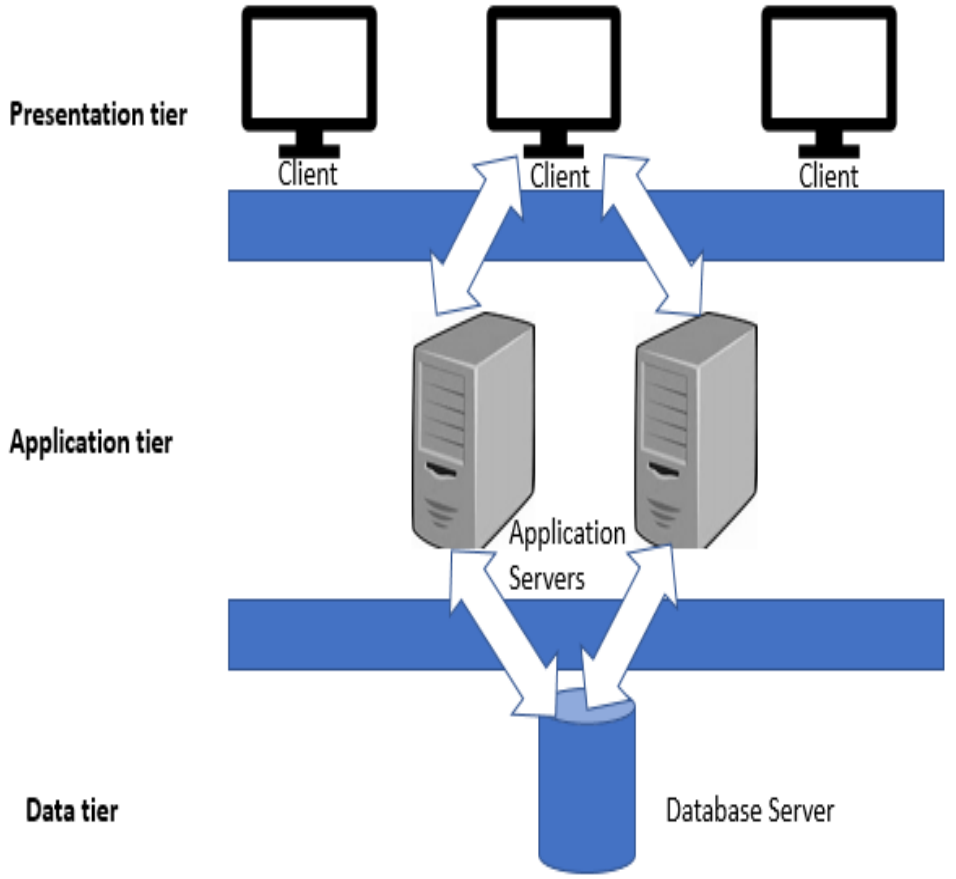
- a server as a shared resource like (a printer, database, or a web server)
- Multiple clients use the shared resource.



Distributed System Architectures:

2- Three-tier:

- clients no longer need to be intelligent
- can rely on a **middle tier** to do the **processing and decision making**.
- Most of the first web applications fall under this category.
- **The middle tier** could be **called an agent** that receives requests from clients, and then forwards it on to the servers.



Distributed System Architectures:

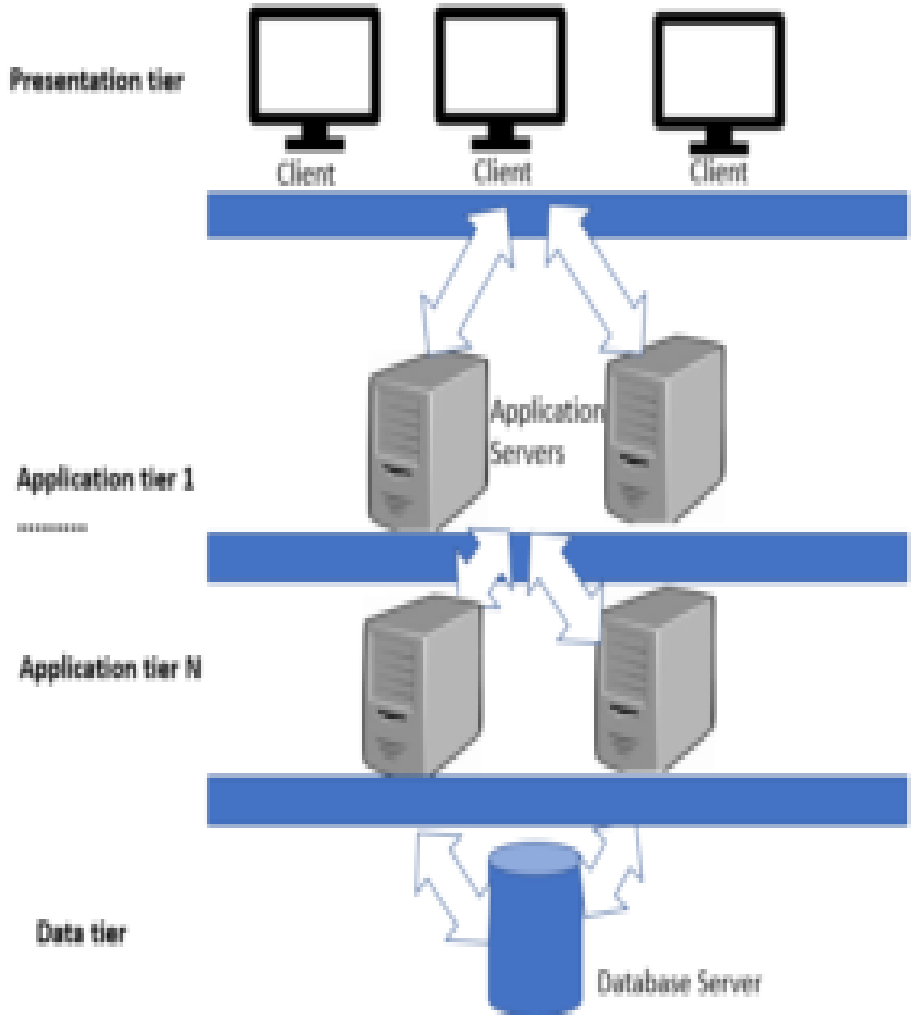
3- Multi-tier:

- Enterprise web services first created **n-tier or multi-tier systems architectures**.

-This popularized the application servers that contain the business logic

- n-tier interacts both with the data tiers and presentation tiers.

- Ex : google.com

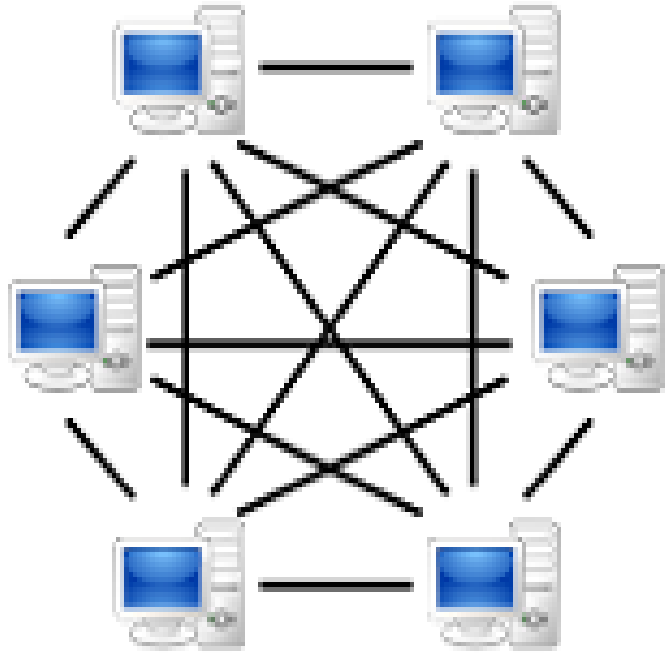


Distributed System Architectures:

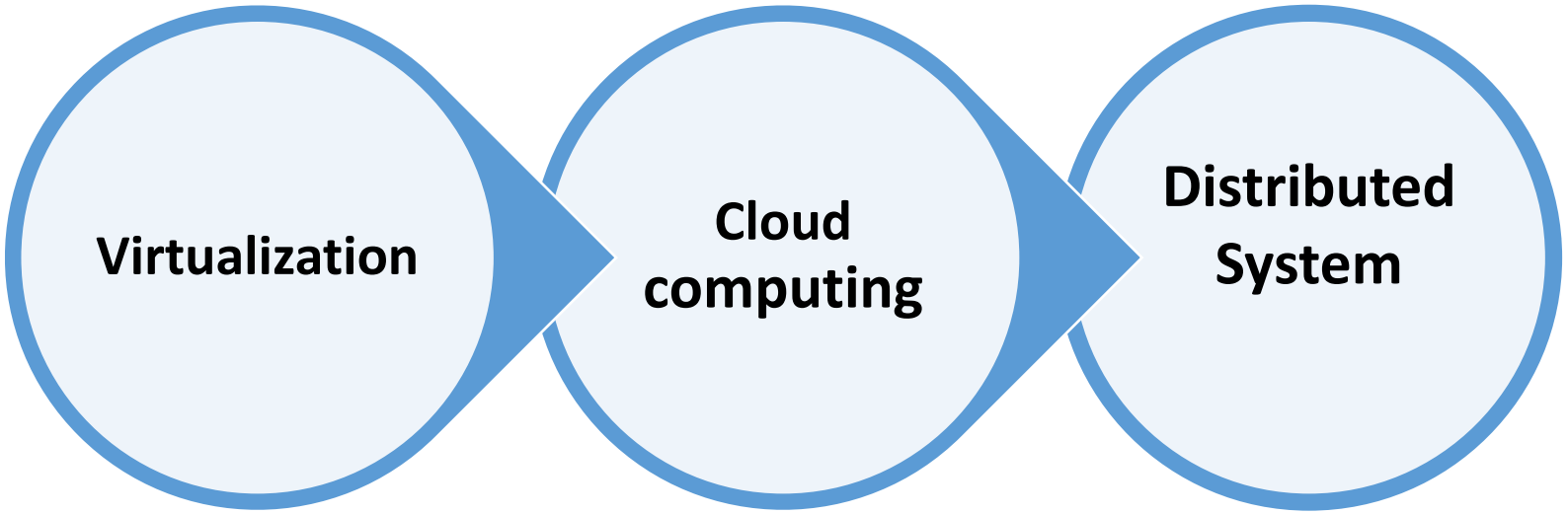
4- Peer-to-peer :

- **No centralized** or special machine that does the heavy lifting and intelligent work in this architecture.

- All the **decision** making and responsibilities are **split up amongst the machines** involved and each could take on client or server roles.
  - Blockchain is a good example of this.



Dependency



Session 3 (Virtualization)

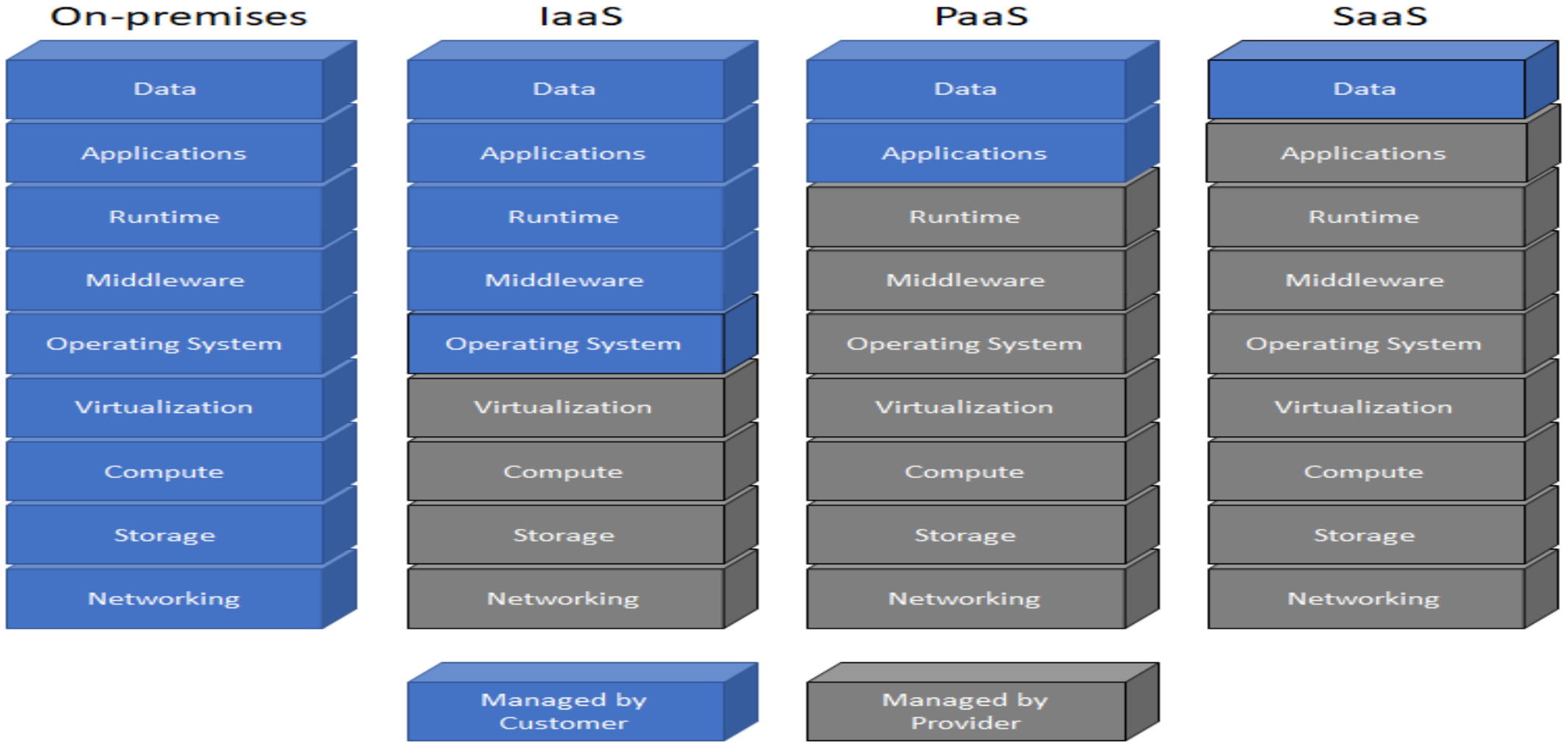
- **Virtualization**
  - Is a technology that run multiple same or different operating systems which is completely isolated from each other
  - Example: run both windows and Linux on the same machine

  - Virtualization is different from Dual Boot ?
  - Dual Boot run only one OS at the same time Virtualization run multiple OS at the same time
- **Virtualization Benefits**
  - Redundancy
  - Legacy hardware
  - Migration
  - Centralized management

Cloud computing

- **Cloud computing**
  - A pool of resources that can be rapidly provisioned in an automated, on-demand manner.
- **Value of cloud computing is :**
  - Economies of scale
  - Elastic enough to scale with the needs of your organization.
  - Cost and operational benefits
  - Easily accessed by users no matter where they reside

Cloud computing service models



Cloud computing service models

- **Software as a service (SaaS).**
  - Customers are provided access to **an application** running on a cloud infrastructure.
  - but the customer has no knowledge of, and does not manage or control, the underlying cloud infrastructure.
- **Platform as a service (PaaS).**
  - Customers can **deploy supported applications** onto the provider's cloud infrastructure,
  - but the customer has no knowledge of, and does not manage or control, the underlying cloud infrastructure.
  - The company owns the deployed applications and data, and it is therefore responsible for the security of those applications and data.
- **Infrastructure as a service (IaaS).**
  - Customers can provision **processing, storage, networks, and other computing resources**, and **deploy and run operating systems and applications.**
  - the customer has no knowledge of, and does not manage or control, the underlying cloud infrastructure. The customer has control over operating systems, storage, and deployed applications, along with some networking components (for example, host firewalls).
  - The company owns the deployed applications and data, and it is therefore responsible for the security of those applications and data.

Cloud computing deployment models:

- **Public.**
  - A cloud infrastructure that is open to use by the general public. It's owned, managed, and operated by a third party (or parties), and it exists on the cloud provider's premises.
- **Community.**
  - A cloud infrastructure that is used exclusively by a specific group of organizations.
- **Private.**
  - A cloud infrastructure that is used exclusively by a single organization. It may be owned, managed, and operated by the organization or a third party (or a combination of both), and it may exist on premises or off premises.
- **Hybrid.**
  - A cloud infrastructure that comprises two or more of the aforementioned deployment models, bound by standardized or proprietary technology that enables data and application portability (for example, fail over to a secondary data center for disaster recovery or content delivery networks across multiple clouds).

Demo on VM _ lab

- Use the VMware Workstation tool to host anther OS on your machine

Thank You